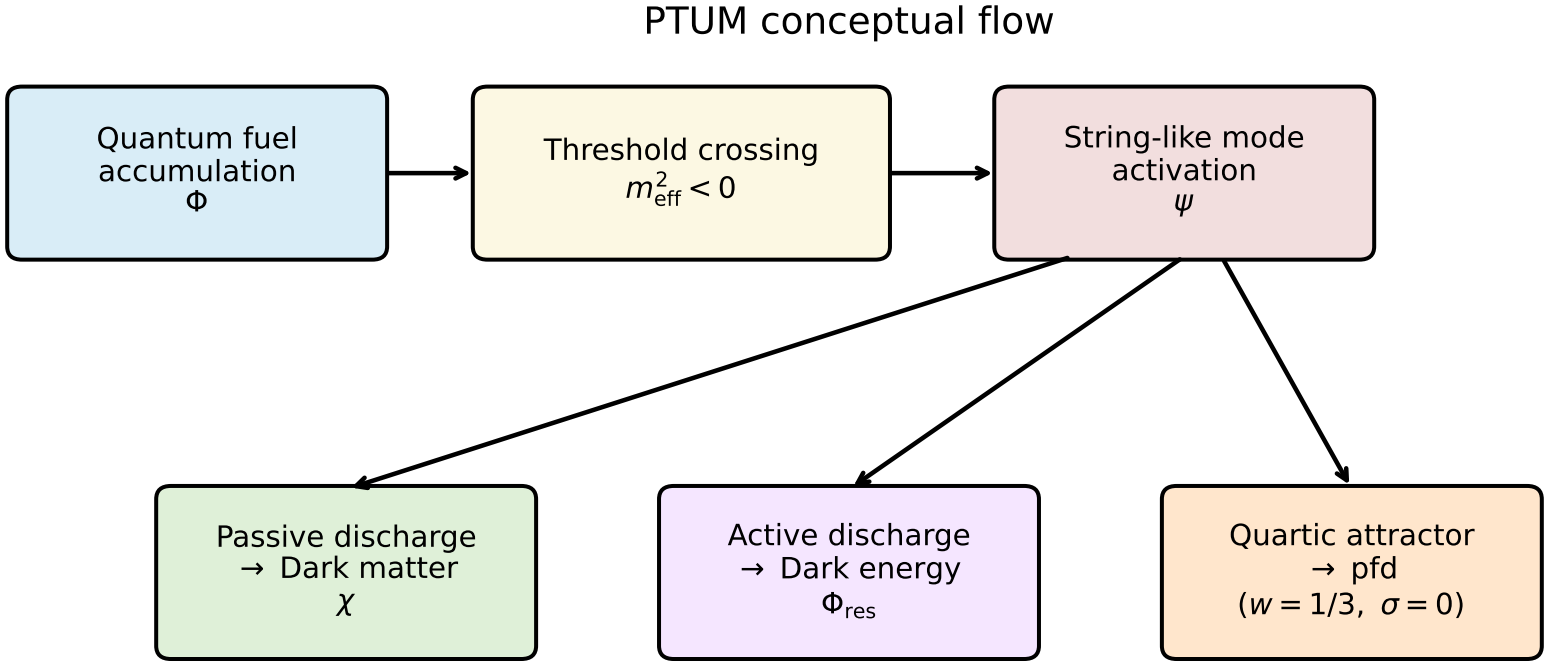
PTUM conceptual flow
Quantum fuel accumulation Φ
Threshold crossing $m_{\text{eff}}^2 < 0$
String-like mode activation ψ
Passive discharge → Dark matter χ
Active discharge → Dark energy Φ_{res}
Quartic attractor → pfd ($w = 1/3,\ \sigma = 0$)